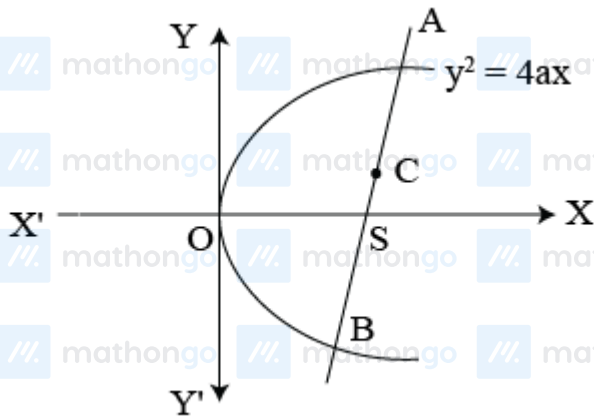


Q1 (1) - Chord and its Properties

If $A(at_1^2, at_1)$, $B(at_2^2, 2at_2)$ is the extremities of a focal chord for the parabola $y^2 = 4ax$, then $t_1 t_2 = -1$



We want to find locus of point $C(\alpha, \beta)$, where C is the centre of the circle having AB as the diameter.

$$\Rightarrow \alpha = \frac{a}{2}(t_1^2 + t_2^2) \text{ and } \beta = a(t_1 + t_2)$$

By eliminating t_1 and t_2 , we get

$$\beta^2 = a^2(t_1^2 + t_2^2 + 2t_1 t_2)$$

$$\Rightarrow \beta^2 = a^2\left(\frac{2\alpha}{a} - 2\right)$$

$$\Rightarrow \beta^2 = 2a(\alpha - a)$$

$$\therefore y^2 = 2a(x - a)$$

Q2 (3) - Chord and its Properties

The length of focal chord making an angle α with the X -axis is $4a \operatorname{cosec}^2 \alpha$. For $\alpha \in [0, \pi/4]$,

its maximum length is $4a \times 2 = 8a$ units.

Q3 (4) - Chord and its Properties

$$\tan \theta = \frac{t_1/2}{t_1^2/4} = \frac{2}{t_1}$$

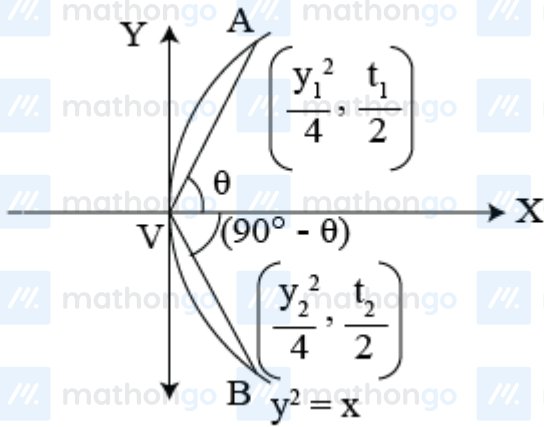
$$\tan(90^\circ - \theta) = \frac{2}{t_2}$$

$$\therefore t_1 = 2 \cot \theta$$

$$\text{and } t_2 = 2 \tan \theta \dots (i)$$

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$$\begin{aligned} \text{Now, } \left| \frac{VA}{VB} \right| &= \frac{\sqrt{\frac{t_1^4}{16} + \frac{t_1^2}{4}}}{\sqrt{\frac{t_2^4}{16} + \frac{t_2^2}{4}}} = \frac{\sqrt{\cot^4 \theta + \cot^2 \theta}}{\sqrt{\tan^4 \theta + \tan^2 \theta}} \\ &= \frac{\cot \theta \cdot \operatorname{cosec} \theta}{\tan \theta \cdot \sec \theta} = \cot^3 \theta \end{aligned}$$

Q4 (1) - Chord and its Properties

Here, $M_{OA} \cdot M_{OB} = -1$

$$\Rightarrow \frac{2}{t_1} \cdot \frac{2}{t_2} = -1$$

$$\Rightarrow t_1 t_2 = -4 \dots (i)$$

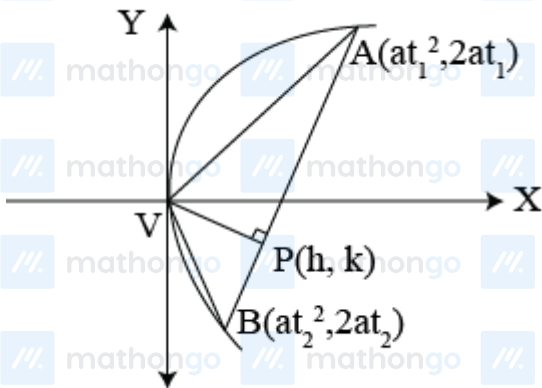
$$M_{OP} \cdot M_{AB} = -1$$

$$\Rightarrow \frac{k}{h} \cdot \frac{2}{t_1 + t_2} = -1$$

$$\Rightarrow t_1 + t_2 = \frac{-2k}{h} \dots (ii)$$

Equation of chord AB , is

$$y(t_1 + t_2) = 2x + 2at_1 t_2, \text{ passes through } (h, k)$$



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$$\Rightarrow k(t_1 + t_2) = 2h + 2at_1t_2 \dots(iii)$$

From Eqs. (i), (ii) and (iii)

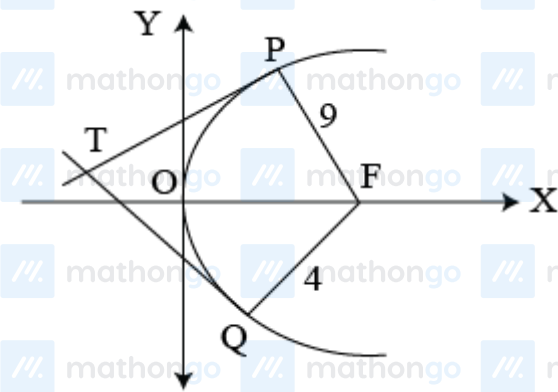
$$k \cdot \left(\frac{-2k}{h} \right) = 2h - 8a$$

$$\Rightarrow -y^2 = x^2 - 4ax$$

or $x^2 + y^2 - 4ax = 0$ is required locus.

Q5 (2) - Chord and its Properties

Since, $(FT)^2 = (FP)(FQ)$



$$\Rightarrow (FT)^2 = 4 \times 9$$

$$FT = 6$$

Q6 (4) - Tangent and its Properties

Given equation of parabola is

$$y = (x - 3)^2 \dots(i)$$

On differentiating Eq. (i), we get

$$\therefore y' = 2(x - 3)$$

Since, tangent is parallel to the chord joining the points (3,0) and (4,1)

$$\therefore 2(x - 3) = \frac{1-0}{4-3}$$

$$\Rightarrow x - 3 = \frac{1}{2} \Rightarrow x = \frac{7}{2}$$

On putting $x = \frac{7}{2}$ in Eq. (i), we get

$$y = \frac{1}{4}$$

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Hence, equation of tangent is $y - \frac{1}{4} = 1 \left(x - \frac{7}{2} \right)$

$$\Rightarrow 4x - 4y = 13$$

Q7 (2) - Tangent and its Properties

The equation of any tangent of slope m to the parabola $y^2 = 4ax$ is $y = mx + \frac{a}{m}$
if it passes through $P(-a, 2a)$, then

$$2a = -ma + \frac{a}{m}$$

$$\Rightarrow 2am = -m^2a + a$$

$$\Rightarrow m^2a + 2am - a = 0$$

Let θ be the angle between the tangents drawn from P . then

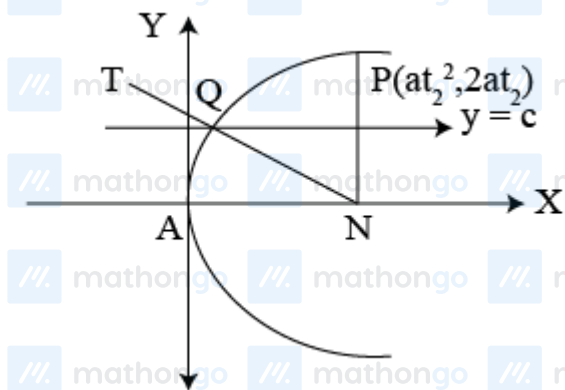
$$\begin{aligned} \tan \theta &= \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} \\ &= \frac{\sqrt{(-2)^2 - 4(-1)}}{1 + (-1)} = \frac{\sqrt{4 + 4}}{0} = \infty \end{aligned}$$

$$\Rightarrow \tan \theta = \infty$$

$$\therefore \theta = \frac{\pi}{2}$$

Q8 (2) - Tangent and its Properties

Equation of PN : $x = at^2$



$y = c$ bisects PN

$$\therefore c = at$$

Which cuts the parabola at Q .

$$\Rightarrow c^2 = 4ax$$

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$$\Rightarrow x = \frac{c^2}{4a}$$

$$\therefore Q\left(\frac{c^2}{4a}, c\right) = Q\left(\frac{at^2}{4}, at\right)$$

$$\therefore \text{Equation of } NQ : y = \frac{-4}{3t}(x - at^2)$$

Which cuts $x = 0$, at $\left(0, \frac{4t}{3}\right)$.

$$\therefore AT = \frac{4at}{3} \text{ and } NP = 2at$$

$$\Rightarrow \frac{AT}{NP} = \frac{4at/3}{2at} = \frac{2}{3}$$

Q9 (2) - Tangent and its Properties

Let the coordinates of B and C be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ respectively.

Then, the coordinates of A are $[at_1t_2, a(t_1 + t_2)]$

The equation of any tangent of $y^2 = 4ax$ is $ty = x + at^2$

$$\therefore p_1 = \frac{at_1t_2 - a(t_1 + t_2)t + at^2}{\sqrt{1+t^2}}$$

$$p_2 = \frac{at_1^2 - 2att_1 + at^2}{\sqrt{1+t^2}}$$

$$\text{and } p_3 = \frac{at_2^2 - 2at_1t_2 + at^2}{\sqrt{1+t^2}}$$

Clearly, $p_2p_3 = p_1^2$

Hence, p_2, p_1, p_3 are in GP.

Q10 (3) - Tangent and its Properties

By definition, $SL = TN$

Q11 (3) - Tangent and its Properties

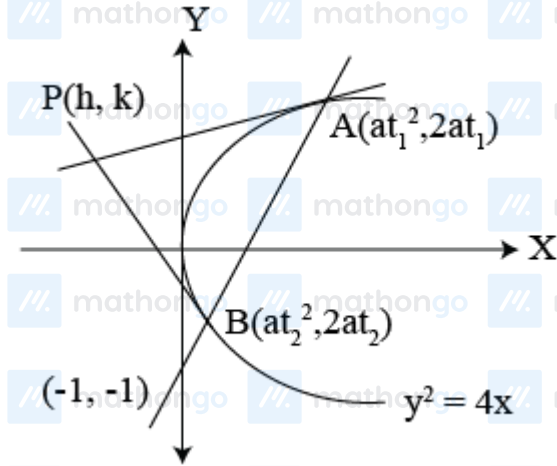
Equation of chord AB , is $y(t_1 + t_2) = 2x + 2at_1t_2$ passing

through $(-1, -1)$

$$\Rightarrow -(t_1 + t_2) = -2 + 2t_1t_2$$

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$$\Rightarrow (t_1 + t_2) + 2t_1t_2 - 2 = 0 \quad \dots (i)$$

$$\text{as, } P(h, k) = P(-t_1t_2, t_1 + t_2)$$

$$\therefore h = -t_1t_2 \text{ and } k = t_1 + t_2 \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\Rightarrow k - 2h - 2 = 0$$

$$\text{or } y = 2(x + 1)$$

Q12 (4) - Tangent and its Properties

The given curve is $y = x^2 + 6$

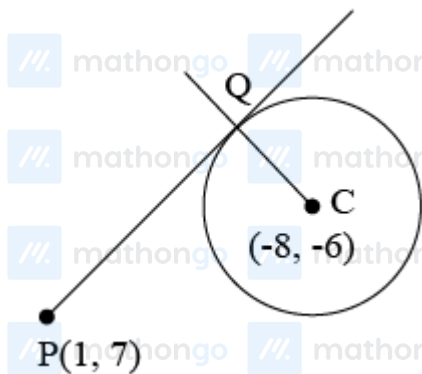
The equation of tangent at (1,7) is

$$\frac{1}{2}(y + 7) = x(1) + 6$$

$$\Rightarrow 2x - y + 5 = 0 \quad \dots (i)$$

According to the question Eq. (i) touches the circle $x^2 + y^2 + 16x + 12y + C = 0$ at Q [centre of circle of

(-8,-6)]



Then, the equation of CQ which is perpendicular to Eq. (i) and passes through (-8,-6) is

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$$y + 6 = \frac{-1}{2}(x + 8)$$

$$\Rightarrow x + 2y + 20 = 0 \dots(ii)$$

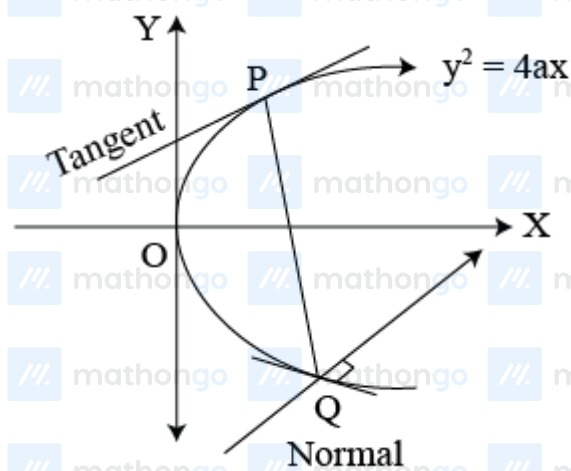
Now, Q is the point of intersection of Eqs. (i) and (ii).

i.e. $x = -6$ and $y = -7$

$\therefore$ The required is point $(-6, -7)$

Q13 (2) - Normal and its Properties

Tangent at P , is



$$y = x + a \dots(i)$$

and equation of normal at Q , is

$$y = x - 2a - a$$

$$\text{or } y = x - 3a \dots(ii)$$

From Eqs. (i) and (ii), distance between parallel lines is $\left| \frac{4a}{\sqrt{2}} \right| = 2\sqrt{2}a$

Q14 (2) - Normal and its Properties

Since, the normal at $(ap^2, 2ap)$ to $y^2 = 4ax$ meets the parabola at $(aq^2, 2aq)$

$$\text{Therefore, } q = -p - \frac{2}{p} \dots(i)$$

Since, $OP \perp OQ$, therefore

$$\frac{2ap-0}{ap^2-0} \times \frac{2aq-0}{aq^2-0} = 1$$

$$\Rightarrow pq = -4$$

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$$\Rightarrow p \left(-p - \frac{2}{p} \right) = -4 \text{ [by Eq.(i)]}$$

$$\therefore p^2 = 2$$

Q15 (3) - Normal and its Properties

Given, a line $y = x - 1$ and a parabola $y^2 = 4x$. On solving these, we have

$$(x - 1)^2 = 4x$$

$$\Rightarrow x^2 - 6x + 1 = 0$$

$$\Rightarrow x = 3 \pm \sqrt{8}$$

$$\text{and } y = 2 \pm \sqrt{8}$$

Suppose point D is (x_3, y_3) . Then,

$$y_1 + y_2 + y_3 = 0$$

$$2 + \sqrt{8} + 2 - \sqrt{8} + y_3 = 0$$

$$\Rightarrow y_3 = -4$$

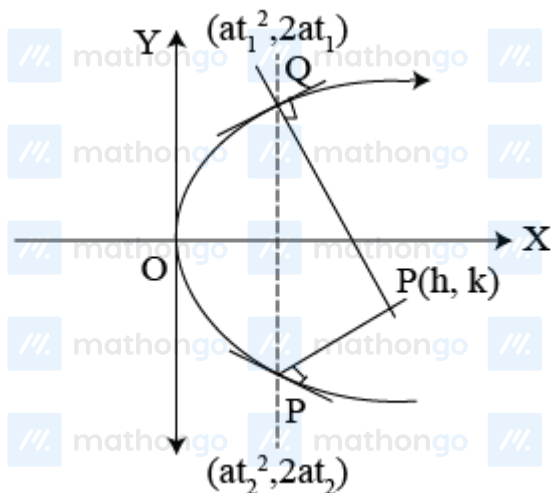
Then $x_3 = 4$

Q16 (1) - Normal and its Properties

$$\text{Here, } m = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2} = \frac{2}{t_2 + t_1} \dots(i)$$

$$\text{and } h = 2a + a(t_1^2 + t_2^2 + t_1 t_2) \dots(ii)$$

$$k = -at_1 t_2 (t_1 + t_2) \dots(iii)$$



From Eqs. (i), (ii) and (iii), we get

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$$h = 2a + a \left[\frac{4}{m^2} + \frac{km}{2a} \right]$$

$$\text{or } 2m^2x = 4am^2 + 8a + ym^3$$

Q17 (2) - Normal and its Properties

Let $P(h, k)$, ($k \neq 0$) through which three normals are drawn to the parabola $y^2 = 4ax$ ($a = 1$) such that product of slopes of two is

$$\alpha_1, \text{ i.e. } m_1m_2 = \alpha.$$

Any normal to the parabola is

$$y = mx - 2am - am^3$$

$$\Rightarrow am^3 + 0m^2 + m(2a - h) + k = 0$$

$$\Rightarrow m^3 + 0m^2 + m(2a - h) + k = 0, \text{ as } a = 1 \dots (i)$$

$$m_1m_2m_3 = -k$$

$$m_3 = \frac{-k}{m_1m_2} = \frac{-k}{\alpha} \text{ [by Eq.(i)]}$$

But m_3 is a root of Eq. (i).

$$\therefore k^3 + (2 - h)k\alpha^2 - k\alpha^3 = 0$$

$$\text{Hence, locus of } P \text{ is } y^2 + (2 - x)\alpha^2 - \alpha^3 = 0 \text{ [} k \neq 0, \text{ cancelled]}$$

Above is same as $y^2 - 4x = 0$, as given.

$$\text{Hence, on comparing } \alpha^2 = 4$$

$$\text{and } 2\alpha^2 - \alpha^3 = 0$$

$$\text{or } \alpha = 0$$

$$\therefore \alpha = 2, \text{ which satisfies both}$$

Q18 (2) - Normal and its Properties

Let $P(at^2, 2at)$ be a point on the parabola $y^2 = 4ax$ such that it cuts the parabola again at $Q(at_1^2, at_1)$.

Then,

$$t_1 = -t - \frac{2}{t}$$

$$\Rightarrow |t| = \left| t + \frac{2}{t} \right|$$

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$$\Rightarrow |t_1| \geq \left| 2\sqrt{t \times \frac{2}{t}} \right| \quad AM \geq C$$

$$\Rightarrow |t| = 2\sqrt{2}$$

$$\text{Now, } OQ = \sqrt{(at_1^2 - 0)^2 + (2at_1 - 0)^2}$$

$$\Rightarrow OQ = at_1 \sqrt{t_1^2 + 4}$$

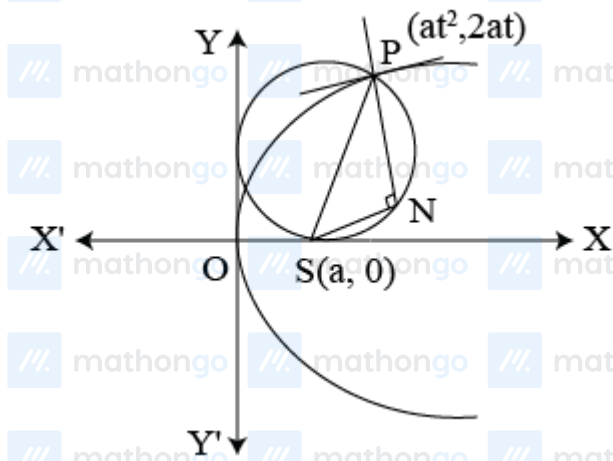
$$\Rightarrow OQ \geq 2a\sqrt{2}\sqrt{8+4}$$

$$\therefore OQ \geq 4\sqrt{6}a$$

Q19 (3) - Normal and its Properties

The equation of normal at $P(at^2, 2at)$ to the parabola $y^2 = 4ax$ is

$$y = -tx + 2at = at^3 \dots (i)$$



Let this normal meets the circle on SP as diameter at N . Then,

$$\angle SNP = \frac{\pi}{2} [\text{an angle in semicircle}]$$

$$\therefore PN^2 = SP^2 - SN^2$$

SN is perpendicular to normal. Now

$$SP = a + at^2$$

$$\text{and } SN = \frac{|at - 2at - at^3|}{\sqrt{1+t^2}} = at\sqrt{1+t^2}$$

$$\therefore PN^2 = a^2(1+t^2)^2 - a^2t^2(1+t^2) = a^2(1+t^2)$$

$$\therefore PN = a\sqrt{1+t^2}$$

Q20 (3) - Normal and its Properties

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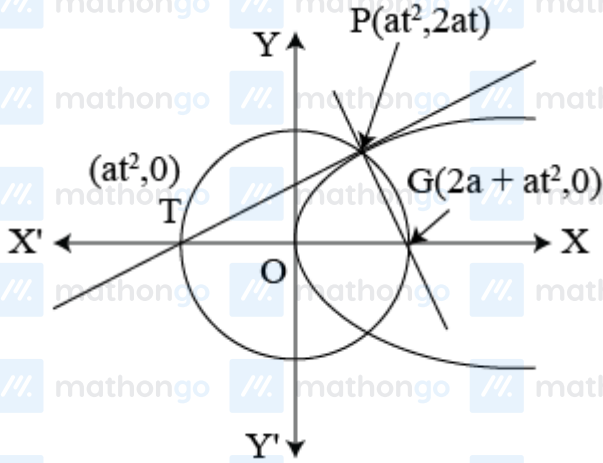
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Tangent and normal at $P(at^2, 2at)$ to the parabola $y^2 = 4ax$ are respectively,

$$ty = x + at^2 \dots(i)$$

$$\text{and } y = -tx + 2at + at^3 \dots(ii)$$



Eqs. (i) and (ii) meet the X -axis, where $y = 0$.

$$\text{From Eq. (i), } x = -at^2$$

$$\text{Therefore, } T \text{ is } (-at^2, 0)$$

$$\text{From Eq. (ii), } tx = 2at + at^3$$

$$\text{Mid-point of } TG \equiv \left(\frac{2a + at^2 - at^2}{2}, 0 \right) \equiv O(a, 0)$$

Since, $\angle TPG = 90^\circ$, the centre of circle through PTG is $(a, 0)$.

If θ is the angle between tangents at P to the parabola and circle through P, T and G , then $(90^\circ - \theta)$ is the angle between PT and OP

$$\text{Slope of } PT = \frac{2at}{2at^2} = \frac{1}{t}$$

$$\text{Slope of } OP = \frac{2at}{a(t^2-1)} = \frac{2t}{t^2-1}$$

$$\text{Now, } \tan(90^\circ - \theta) = \frac{\left| \frac{1}{t} - \frac{2t}{t^2-1} \right|}{\left| 1 + \frac{1}{t} \left(\frac{2t}{t^2-1} \right) \right|} = \frac{1}{t}$$

$$\Rightarrow \cot \theta = \frac{1}{t} \Rightarrow \tan \theta = t$$

$$\therefore \theta = \tan^{-1} t$$

Q21 (3) - Normal and its Properties

$$(c) t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_1 t_2 = -t_1^2 - 2$$

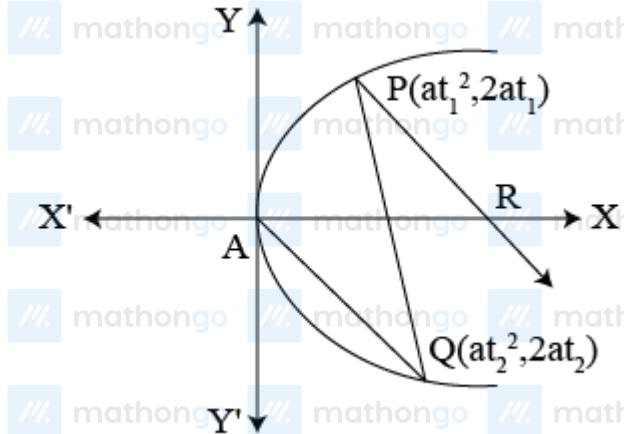
The equation of a line through P parallel to AQ is

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$$y - 2at_1 = \frac{2}{t_2}(x - at_1^2)$$



$$\begin{aligned} \text{Put } y = 0, \text{ then } x &= at_1^2 + at_1t_2 = at_1^2 + a(-2 - t_1^2) \\ &= 2a + 2a_1^2 = 2(a + at_1^2) \end{aligned}$$

which is twice the focal distance of P .

Q22 (1) - Miscellaneous

We know that the locus of the point of intersection of tangents to $y^2 = 4ax$ which meet at an angle α is given by

$$y^2 - 4ax = (a + x)^2 \tan^2 \alpha \dots (i)$$

$$\text{Here, } \alpha = 45^\circ$$

Putting $\alpha = 45^\circ$ in Eq. (i), we get

$$y^2 - 4ax = (a + x)^2$$

Hence, the required locus is

$$y^2 - 4ax = (a + x)^2$$

Q23 (4) - Miscellaneous

Equation of the chord of contact of tangents drawn from a point (3,4) to the parabola $y^2 = 2x$ is

$$yy_1 = 2(x + x_1)$$

$$\Rightarrow y(4) = 2 \times \frac{1}{2}(x + 3)$$

$$\Rightarrow 4y = x + 3$$

$$\therefore x - 4y + 3 = 0$$

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Q24 (1) - Miscellaneous

Let the tangents intersect at $P(h, k)$. Then, $lx + my + n = 0$ will be chord of contact of P . That means

$lx + my + n = 0$ and $yk - 2ax - 2ah = 0$ will represent the same line.

$$\text{Thus, } \frac{k}{m} = \frac{-2a}{l} = \frac{-2ah}{n}$$

$$\Rightarrow h = \frac{n}{l}, k = \frac{-2am}{l}$$

Q25 (1) - Miscellaneous

If (h, k) be the point of intersection of tangents, then

$$ky = 2a(x + h) \dots (i)$$

is same as normal

$$y = mx - 2am - am^3 \dots (ii)$$

On comparing Eqs. (i) and (ii), we get

$$m = \frac{2a}{k}, -2am - am^3 = 2a \cdot \frac{h}{k}$$

On eliminating the variable m between the above two relations, we get

$$-2a \left(\frac{2a}{k} \right) - a \cdot \frac{8a^3}{k^3} = 2a \cdot \frac{h}{k}$$

$$\Rightarrow h + 2a + \frac{4a^3}{k^2} = 0$$

$$\Rightarrow k^2(h + 2a) + 4a^3 = 0$$

Hence, the required locus in

$$y^2(x + 2a) + 4a^3 = 0$$

Q26 (1) - Miscellaneous

Let the pole be (x_1, y_1) . Then, the equation of polar with respect to $y^2 = 4ax$ is

$$yy_1 = 2a(x + x_1)$$

$$\Rightarrow 2ax - yy_1 + 2ax_1 = 0 \dots (i)$$

Clearly, Eq. (i) and $lx + my + n = 0$ represent the same line

$$\therefore \frac{2a}{l} = \frac{-y_1}{m} = \frac{2ax_1}{n} \Rightarrow x_1 = \frac{n}{l}$$

$$\text{and } y_1 = \frac{-2am}{l}$$

Hence, the pole of given line is $\left(\frac{n}{l}, \frac{-2am}{l} \right)$.

Basic Level DPP - Parabola

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